

HunterR Map Editor

Complete User Guide

Version 0.4.2

A full-featured terrain and track mod editor for the Railroader train simulation game. Edit heightmaps and vegetation, build and modify track layouts, manage progression systems, place scenery prefabs, and hot-reload every change directly into a running game.

Covers: Setup · Terrain · Track · Geo Tools · Splineys · Mod Panels · Mandela · Calculators ·
TrackBridge

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1. Getting Started

1.1 Requirements

Python	3.10 or newer. Python 3.14 is recommended. Download from python.org.
pip libs	Install with: <code>pip install pygame pillow numpy</code>
Game	Railroader installed via Steam. Tile .data files come from the game install.
Mod folder	An existing Railroader mod folder containing Definition.json, or create a new one inside the editor.

1.2 Launching

Extract the editor zip to any folder on your machine. Then run:

```
python -m edit_tiles
```

On Windows, double-click **run_editor.bat**. To pre-load tile folders, pass them as arguments:

```
python -m edit_tiles C:\path\to\tiles D:\more\tiles
```

Tip: You can also drag-drop any tile folder directly onto the editor window after it opens.

1.3 The Interface

The window has two button rows at the top, a main map view, and a status bar at the bottom. The top-left corner shows the properties panel when a node or segment is selected.

Row 1 — Terrain	Heightmap · Vegetation · Water · Hillshade · Fit · Tracks · Nodes · Load Graph · Load Tiles · Reload
Row 2 — Mod	●LIVE · Generate · Mod · Prog · Areas · Spans · Scenery · Group · Calc · Mandela · Spliney · Geo · OSM · Export · Diff · ? · Edit
Status bar	Every action prints feedback here. If something seems broken, read this first.
Minimap	Top-right corner — full overview of all loaded tiles at all times.
Properties panel	Top-left — appears when a node or segment is selected. All fields are editable.

1.4 Quick Start — First time workflow

1. Click **Load Tiles** and select your tile folder. The terrain map will appear.
2. Click **Mod** → **Open Mod Folder** and select your mod directory.
3. Track appears on the map. Click any segment or node to select it.
4. Use **Ctrl+click** on empty map space to place new nodes.
5. Select a node, click **Connect** → in the properties panel, then **Ctrl+click** another node to draw a segment between them.
6. All changes auto-save and hot-reload into the running game in about 500ms (requires TrackBridge mod installed in-game).

Note: The editor always writes to game-graph.json in your mod folder. Ctrl+S force-saves everything. Ctrl+Z undoes up to 50 mod edits.

2. Navigating the Map

2.1 Mouse Controls

Scroll wheel	Zoom in and out	Zoom in far	Node IDs and labels appear at high zoom
LMB drag	Pan the map	Hover tile	I key toggles tile info tooltip (coords, height range, veg, water)
MMB drag	Pan the map (middle mouse button)	Minimap	Always visible top-right — shows the full extent
F	Fit all loaded tiles into view	Status bar	Bottom of window — shows current action and feedback
R	Redraw / refresh the view	Ctrl+- / =	Shrink or grow the entire UI scale
Drag-drop	Drop a tile folder onto the window to load it	Ctrl+0	Reset UI scale to 100%

2.2 Display Modes

Heightmap (H)	Primary view. Elevation shown as colour-graded contours. Dark green = low, bright green = mid, yellow/white = high. Press S to toggle directional hillshade lighting.
Vegetation (V)	Shows dominant biome/vegetation type per pixel. 7 presets (0–7) representing different terrain types. In edit mode, press 0–7 to paint vegetation.
Water (W)	Water coverage mask. White = water, black = land.
Diff (D)	Highlights tiles modified since last save with a yellow border. Use this to check what will be written before Ctrl+S.
Hillshade (S)	Toggles directional lighting over any display mode. Makes terrain shape and slopes much easier to read.
Tracks (T)	Toggle the entire track overlay on and off.
Nodes (N)	Toggle node dot rendering. Useful at high zoom to reduce clutter.

2.3 Track Overlay Colors (with mod loaded)

Track segments are colored by class AND style:

Yellow	Mainline class, Standard style
Orange	Branch class, Standard style
Purple	Industrial class, Standard style
Brown	Any class with Yard style
Blue	Any class with Bridge (Trestle) style
Red	Any class with Tunnel style
Steel blue	Base-game reference track (read-only, from game-graph layer)
Blue polylines	Rivers (FlowyThingBuilder splineys)
Tan polylines	Roads (FlowyThingBuilder splineys)

2.4 OSM Overlay (O key or OSM button)

Toggles an OpenStreetMap tile overlay over the terrain. Useful as a reference for placing roads, rivers, and track to match real geography. The OSM zoom level can be nudged with the OSM panel controls.

Note: OSM tiles are fetched from the internet on first view. They are cached locally so panning over the same area is fast on subsequent sessions.

3. Terrain Editing

Press **E** to toggle terrain edit mode. The cursor becomes a brush circle. Left-click paints, right-click erases/lowers. A toolbar strip appears below the nav bar with brush controls.

3.1 Brush Controls

E	Toggle edit mode on / off	Raise	Raise terrain (LMB) or lower (RMB) smoothly based on strength
B	Cycle through brush modes	Flatten	Level the painted area to the height you first clicked
[]	Decrease / increase brush size	Paint	Stamp an exact target height value without blending
Ctrl+Scroll	Brush size via mouse wheel	Smooth	Average pixels with their neighbours — softens sharp edges
- =	Decrease / increase brush strength	Noise	Add fractal Brownian motion procedural texture
MMB	Eyedropper — sample height or veg at cursor	Erode	Simulate hydraulic erosion for natural-looking terrain
LMB	Paint / raise terrain	,	Set height floor clamp to cursor height (blocks lowering below)
RMB	Erase / lower terrain	.	Set height ceiling clamp to cursor height (blocks raising above)
Ctrl+Z	Undo last brush stroke	Ctrl+,	Clear the height floor clamp
Ctrl+S	Save all modified tiles to disk	Ctrl+.	Clear the height ceiling clamp

3.2 Vegetation Painting

Switch to Vegetation mode (V) while in edit mode. Press **0–7** to select a vegetation preset, then paint with LMB. The Noise brush creates varied natural-looking biome patches.

3.3 Selection Tools (M in edit mode)

M	Toggle selection mode on / off	Use case	Selection is useful for copying terrain features between areas or exporting a region as a reference image
Rect	Click and drag a rectangle to select		
Lasso	Freehand click-drag selection		
Wand	Click to flood-fill select pixels with similar values	Paste mode	After Ctrl+V, move the mouse to position the pasted region, then click to commit
Ctrl+C	Copy the selected region	Wand range	The wand selects pixels within a similarity tolerance of the clicked value
Ctrl+V	Paste — click to place the copied region anywhere		
Ctrl+X	Cut the region (also exports as PNG)	PNG export	Ctrl+X exports the selected region as a PNG height image
Delete	Clear the selected region to zero		

3.4 Saving Terrain

Modified tiles get a yellow border in Diff mode. Press **Ctrl+S** to save all dirty tiles. Tiles are saved as PNG data in RGBA format inside the .data files — the R and G channels encode the 16-bit height value, A encodes vegetation.

Important: Terrain undo only works within the current session. Closing the editor without saving will lose all unsaved paint work.

3.5 Tile Generator (Generate button)

The Generate panel downloads real-world elevation and land-cover data from Mapbox and converts it into editor-compatible tile files. You need a free Mapbox API token from mapbox.com to use this feature.

Token field	Paste your Mapbox public token here, or use the Paste button to pull from clipboard
Output folder	Where generated tiles will be saved
NLCD	Use NLCD land-cover data for vegetation (USA only)
NLCD blur	Blur radius to smooth biome boundaries
Veg override	Force a single vegetation preset for all generated tiles
Workers	Number of parallel download threads (default 4)
Grid	Click cells to queue them. Drag to box-select. RMB to dequeue.
Run	Start downloading and generating all queued tiles
Progress	Each queued cell shows its download/generation status
Presets	Save and load your token/output/settings as named presets

Note: Generated tiles use the same .data format as hand-edited tiles. You can mix generated and hand-edited tiles freely.

4. Loading a Mod Project

Click the **Mod** button to open the mod manager panel. A mod project is a folder containing a Definition.json file and one or more JSON layer files.

4.1 Opening a Mod

Open Mod Folder	Select the root folder of an existing mod. Must contain Definition.json.
Open Base Game	Load a standalone graph-data.json or game-graph.json file directly. Useful for viewing base-game track without a full mod context.
New Mod	Creates a new mod folder from scratch. The editor prompts for a mod ID and display name, then creates the required folder structure and Definition.json automatically.

4.2 The Layer List

Once a mod is loaded, the Mod panel shows all JSON layers it contains. Each layer is a separate file (e.g. graph-data.json, splineys.json).

Layer dot	Click to toggle this layer's visibility on the map	Active layer	New nodes, segments, and splineys are always written to the active layer
Layer row	Click to make this the active write layer for new edits	Save All	Ctrl+S saves all layers with unsaved changes
● indicator	This layer has unsaved changes	Reload	Ctrl+R reloads all layer data from disk (yellow = unsaved changes will be lost)
Nodes col	Node count + deleted node count	Escape	Closes the panel but the mod stays loaded
Segs col	Segment count for this layer	Multiple layers	You can load a mod with many layers and toggle each on/off for clarity
Spl col	Spliney (road/river) count		
Areas col	Area/town count in this layer		

4.3 What the Map Shows

Yellow lines	Track segments in your mod (Mainline class, Standard style)
Orange lines	Branch class segments
Purple lines	Industrial class segments
Brown lines	Yard-style segments
Blue lines	Bridge/trestle-style segments
Red lines	Tunnel-style segments
Steel blue lines	Base-game reference track (read-only)
Blue polylines	Rivers
Tan polylines	Roads
Grey/cream blobs	AutoTrestle bridge geometry

Orange circles	Turntables
Green diamonds	Station areas
Orange diamonds	Loaders and industry areas
Coloured squares	Scenery/prefab objects (zoom in for model name label)
Large circles	Area/town centres with name label
Orange dot	Live locomotive position (requires TrackBridge)
Teal dot	Live railcar position (requires TrackBridge)

5. Track Editing

5.1 Selecting Nodes and Segments

Click anywhere on a track line to select the nearest segment. Click a node dot to select the node. The properties panel appears in the top-left corner immediately.

Click segment	Select — click anywhere along the line	Node label	Node ID shown next to the dot at high zoom
Click node	Select the node	Connected segs	Listed at the bottom of the node properties panel — clickable
Click empty space	Deselect everything	Segment midpoint	Grade % label shown at midpoint when grade labels are enabled
Click selected node	Second click starts drag mode	Select from list	Click a segment ID in the node panel to select that segment
Drag release	Commit move — Y auto-sampled from terrain	Undo	Ctrl+Z — up to 50 steps for all mod edits
Escape	Cancel drag, connect mode, or placement	Auto-save	Every edit auto-saves and triggers bridge hot-reload
Delete key	Delete the selected node and all its segments		

5.2 Creating Nodes

Ctrl+click map	Place a new node at the cursor. Y is auto-sampled from terrain under the cursor.
Geo → Add Node	Enter placement mode. A crosshair shows live X/Z coordinates as you move. Click to place, Escape to cancel. Supports Y-lock and Y-inherit modes.
Y lock mode	In Add Node mode: enable Place Y Lock to pin all new nodes to a fixed elevation.
Y inherit mode	In Add Node mode: new nodes inherit the Y of the last placed node. Useful for flat yard tracks.

5.3 Connecting Nodes

Connect → button	In the node properties panel. Enters connection mode from the selected node.
Ctrl+click node	While in connect mode, connects to the clicked node with a new segment.
Drag node → node	A snap ring appears when you drag near another node. Release to create a connecting segment.
Drag node → seg	A cyan highlight appears on the segment. Release to insert the dragged node into that segment. The original segment is replaced by two new segments.
Shift+drag → seg	Yellow highlight. Inserts the node AND adds a turnout diverge leg using the current Geo → Turnout settings.

Tip: When you insert a node into a segment, the new node automatically gets the bezier tangent heading at the insertion point — it flows smoothly with the curve.

5.4 Moving Nodes

Click a node once to select it. Click it a second time to enter drag mode — the cursor changes. Move to the new position and click again (or release) to commit. The Y position is automatically sampled from the terrain under the new position. Press Escape at any time to cancel the drag.

6. Node & Segment Properties

6.1 Node Properties Panel

Select any node to open this panel. All fields are click-to-edit — click a field, type the new value, press Enter to commit.

X / Y / Z	Direct 3D position editing. Y is elevation in metres.
RotY nudge	Fine-tune the node heading with buttons: ± 90 / 45 / 30 / 15 / 10 / 5 / 1 / 0.1 / 0.05 / 0.001 degrees. The heading determines the bezier tangent direction for all connected segments.
Flatten	Zeros rotX and rotZ — levels the node so it sits flat on the terrain. Use after placing nodes on steep slopes.
Reverse	Flips the node heading 180 degrees. Useful when a switch stand faces the wrong way.
Merge	Only active when the node has exactly 2 connected segments. Removes the node and joins its two segments into one.
Split	Duplicates the node and rewires the currently selected segment to the copy. Creates two nodes at the same position — drag one away.
Flip Stand	Toggle the flipSwitchStand flag — moves the switch stand to the other rail side.
Copy XYZ	Copies the node X/Y/Z position to the system clipboard.
Paste Y	Pastes only the Y (elevation) value from the clipboard onto this node.
Connected segs	Clickable list of all segments touching this node. Click any segment ID to select that segment.

6.2 Segment Properties Panel

Track Class	Mainline / Branch / Industrial — one-click colour buttons. Controls default speed, AI routing cost, and display colour.
Style	Standard / Yard / Bridge / Tunnel — one-click toggle. Controls visual appearance and segment colour on the map.
Speed limit	Speed in mph. 0 means use the class default (Mainline=45, Branch=25, Industrial=15). Use nudge buttons ± 25 /10/5/1 or click the value box to type.
Priority	Integer from -2 to +2. Higher = preferred by AI route search. Use +1 or +2 on through routes to guide AI past turnouts correctly.
GroupID	Assigns this segment to a track group. Groups are unlocked by the progression system. Leave blank for always-available track.
Reverse	Swaps startId and endId. Changes which direction the segment 'starts' from.
Trestle	Wraps this segment in an AutoTrestleBuilder spline entry, adding bridge timber visuals over the segment automatically.

Tip: The Track Class buttons brighten to show the active class. After changing class, the style buttons reset to show what is valid for that class.

6.3 Track Classes Explained

Mainline	45 mph default. Used for main through routes. Yellow on map (Standard style).
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Branch	25 mph default. Secondary lines, branch routes, turnout through routes. Orange on map.
Industrial	15 mph default. Yard tracks, spurs, diverging legs, slow service tracks. Purple on map.

7. Geo Tools

Click **Geo** to open the Geometry Tools panel. All tools show an orange ghost preview on the map before any data is written. Use Preview to check, Commit to write, Clear to discard.

7.1 Arc Tool

Generates a constant-radius curve from the selected node. Select a node first, then configure and preview.

Radius (m)	Curve radius. Larger = gentler curve. 60 m is the safe minimum — the editor shows a warning below this.
Arc (degrees)	Total sweep angle. 90° = quarter circle, 180° = half circle.
Height change	Optional rise or fall in metres over the full arc length.
Segments	0 = auto-calculated. Enter a number to force a specific segment count.
Speed limit	Applied to all segments in the arc.
Direction	Left or Right — relative to the direction of travel from the anchor node.
Class / Style	Applied to all generated segments.

Note: A red 'Blocked' message appears if the radius is below 60 m. Unsafe geometry cannot be committed.

7.2 Parallel Tool

Creates one or more parallel copies of a selected segment chain, offset laterally by a fixed distance. Used for building double track, sidings, and yard ladders.

Source	The selected segment or grade chain to copy
Offset (m)	Lateral distance between the original and the copy
Side	Left, Right, or Both
N tracks	Number of parallel copies to generate
Class	Track class for the copied segments

7.3 Fit Arc Tool

Solves a constant-radius arc that best fits the current grade chain. Build a grade chain first (Grade tool, or select a chain of connected nodes), then click Fit Arc to solve the circle.

RMS error	Reported after solving — lower is a tighter fit to your chain
Radius result	The solved radius. If below 60 m, the commit is blocked.
Turnout block	Fit Arc refuses to run if the chain contains any switch/frog nodes
Preview	Shows the fitted arc in orange on the map
Commit	Replaces the chain nodes with the fitted arc geometry

7.4 Add Node Tool

Places new nodes directly from the Geo panel with precise control over Y positioning. Three Y modes are available:

Terrain Y	Default. New node Y is sampled from the terrain height at the click point.
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Lock Y	All new nodes are placed at a fixed Y elevation that you set. Good for flat yard areas and bridges.
Inherit Y	Each new node gets the Y of the previously placed node. Good for building a level section step by step.

7.5 Grade Tool

Adjusts node elevations (Y values) across a chain to achieve a target grade profile. Build the chain first by selecting nodes in order.

Grade Set Start	Sets the first node of the grade chain from the selected node.
Grade Add Node	Appends the selected node to the end of the chain.
Fix First	Lock the start node's elevation — only intermediate nodes move.
Fix Last	Lock the end node's elevation.
Smooth Grade	Redistributes elevation evenly across the chain by arc distance.
Apply Grade %	Forces a constant percentage grade from start to end. Enter the target grade (e.g. 2.5 for 2.5% uphill).
Straighten XZ	Keeps all elevations but snaps X/Z positions to a straight line between start and end. Useful after grade smoothing on curves.

Tip: A 2.5% grade is comfortable. Anything above 4% is steep. Above 6% is very steep and some locos may stall on it.

8. Turnout, Wye & Curved Turnout Tools

The Turnout and Wye tabs in the Geo panel build complete switches from a selected frog node. The game engine requires exactly 3 segment connections per switch node — the editor enforces this automatically.

8.1 Standard Turnout

Select the frog node (the switch point), open Geo → Turnout, configure, Preview, Commit.

Diverge angle	How many degrees the branch leg diverges from the approach. No.6 frog = 9.46°, No.8 = 7.13°, No.10 = 5.71°, No.12 = 4.76°.
Leg length (m)	Length of each exit leg from the frog. Longer = wider spread at the ends.
Main speed	Speed limit for the through route.
Diverge speed	Speed limit for the branch leg (usually lower than main).
Direction	Left or Right — which side the branch leg goes.
Main class	Track class for the through route.
Div class	Track class for the diverging branch.
Flip stand	Moves the switch stand to the opposite rail side.
Through curve °	0 = straight through (standard turnout). Non-zero = curved turnout. See section 8.3 below.
Safe checks	Blocked if leg < 18 m, angle > 15°, or diverge radius < 60 m.

8.2 Turnout Templates

Save and reload full turnout configurations. Templates are stored in `turnout_templates.json` in the editor folder.

Load...	Opens a searchable dropdown of all saved templates. Click one to load its settings.
Save as...	Saves the current settings under a name you type.
Delete	Removes the currently loaded template. Only enabled when a template is active.

Built-in templates:

No. 6 Yard	9.46° / 20 m / Branch class — compact yard use
No. 6 Industrial	9.46° / 18 m / Industrial class — spur service
No. 8 Branch	7.13° / 26 m / Branch class — secondary lines
No. 8 Mainline	7.13° / 30 m / Mainline class — general purpose mainline
No. 10 Mainline	5.71° / 38 m / Mainline class — moderate speed
No. 11 Mainline	5.19° / 44 m / 45 mph — high-speed branch
No. 12 High Speed	4.76° / 50 m / 45 mph — high-speed mainline diverge
Curved No.6 R150	Curved turnout designed for a 150 m radius main line
Curved No.8 R300	Curved turnout designed for a 300 m radius main line
Curved No.10 R400	Curved turnout designed for a 400 m radius main line

8.3 Curved Turnout

A curved turnout is placed on a section of track that is already on a curve. Both the through leg and the diverge leg arc away from the frog — neither leg is straight. Set **Through curve °** to a non-zero value to enable this. The editor will auto-calculate the correct value from the curvature of the connected entry segment when you hit Preview.

Through curve °	How many degrees the through leg curves. Leave at 0 for a standard straight-through turnout. Set positive for a curved turnout (same direction as the diverge branch). Auto-calculated from the connected segment's curvature rate when set to 0.
Diverge angle	Total angle of the branch from the approach — must be greater than Through curve °.
Auto-calc	With Through curve = 0, the editor measures the heading change per metre of the entry segment and multiplies by leg length to derive the through curve. The status bar shows the computed value as 'curve: auto X.XX°'.

8.4 Wye Tool

A wye switch has one entry and two diverging legs — no through route. Used where a line splits symmetrically into two branches, or for run-around moves.

Left angle	Degrees the left exit leg diverges from the approach direction.
Right angle	Degrees the right exit leg diverges from the approach direction.
Leg length	Length of each exit leg from the frog.
Class / Style	Applied to all three segments of the wye.
Flip stand	Flip the switch stand side.

Note: The frog node gets exactly 3 connections (entry + left + right). This satisfies the game engine's hard requirement of exactly 3 segments per switch node.

9. Pieces Tool — Model Railroad Mode

The Pieces tool lets you build track section by section, exactly like snapping model railroad track together. Each piece extends from the endpoint of the previous one. When you are happy with the layout, click **Bake Pieces** to write everything at once.

9.1 Workflow

1. Select node	Click any existing node on the map to use as the starting anchor.
2. Set Start	Click Set Start in the Geo → Pieces panel. The chain begins here.
3. Choose type	Pick Straight, Arc, or Turnout from the Piece type selector.
4. Set params	Straight: set length. Arc: set radius, angle, and direction. Turnout: uses current Geo → Turnout settings.
5. Add Piece	Click Add Piece to append to the chain. Preview updates on the map.
6. Repeat	Keep adding pieces. Each one starts from the end of the previous one.
7. Bake Pieces	Click the large green button to write all nodes and segments permanently.

9.2 Piece Types

Straight	Set Length (m). Continues at the current heading.	Chain list	Last 6 pieces shown at the bottom of the panel.
Arc	Set Radius (m), Angle (deg), and Left/Right turn direction.	End pose	Shows current X/Z and rotY at the tip of the chain.
Turnout	Uses current Geo→Turnout settings (leg length, diverge angle, direction, flip, div class).	Draft total	Running count of pieces, nodes, segments, and total length in metres.
Class	Set once and all new pieces use it until you change it.	Undo Last	Removes the most recent piece from the chain.
Style	Standard / Yard / Bridge / Tunnel per piece.	Clear Draft	Resets everything including the anchor.
Speed	Speed limit per piece.	Bake	Large green button — only active when a preview exists.

9.3 Tips

Tip: Yard ladder: Set Start on an existing node, then alternate Straight pieces (for clearance) and Turnout pieces (alternating Left/Right direction). Hit Bake when the ladder looks right.

Tip: For double-track entry, use two Pieces chains: one going left-turnout, one going right-turnout from the same anchor node, then Bake each one separately.

Tip: On a curved chain, turnout through-curve is auto-calculated from the curvature of the previous piece. No manual entry needed.

10. Spliney Editor — Roads, Rivers & Trestles

Splineys are FlowyThingBuilder entries that create road and river visuals in Railroader. They are stored as a list of control points with position, rotation, and width. The game uses them to generate mesh geometry along the path.

10.1 Spliney Panel (Spliney button)

Type	Road or River	Splineys list	All FlowyThing entries in all loaded layers
Target	Pick the writable JSON layer for new splineys	Details card	Shows style, width, point count, and total length
New JSON...	Create and register a new road/river JSON file as a new layer	Go To	Pan map to the selected spliney's first point
Width	Default width for new placement	Delete	Remove the selected road/river spliney
Seed heading	Starting direction for the first two points	Writes to	Chosen target JSON layer, then bridge hot-reload
Use Sel Rot	Use selected track node heading for the seed direction		
Place on Map	First click drops a new spliney with 2 starter points		

10.2 Editing Control Points

Zoom in on the map until yellow control point dots become visible. Click a dot to select it. Click again to enter drag mode.

Click dot	Select the control point	Ins Before	Insert a new point before the selected one
Click again	Enter drag mode	Ins After	Insert a new point after the selected one
Drag release	Move the point; Y is optionally resampled from terrain	Prev / Next	Walk through points without re-clicking on the map
Sample Y	Resample this point's height from terrain	Fit Terrain	Resample all point heights to the terrain in one pass
Auto Rot	Recalculate rotY from neighboring points	Reverse Flow	River only: reverse point order and add 180° to rotY
Delete Pt	Remove this point (spline must keep at least 2)	Flow arrows	Map preview showing river direction (first → last point)
Width field	Per-point width editable in the point inspector	Trestle	Segment panel — wraps a track segment in AutoTrestleBuilder

10.3 Geo → Spliney Trace Workflow

For tracing a road or river from map contours, use the Spliney tab in the Geo panel. This lets you sketch a guide path first, then convert it to a spliney.

Trace	Click map to add guide points one at a time
Stop Trace / RMB	Exit guide tracing

Use Chain	Copy the current grade chain or selected segment chain as the guide
Undo Pt	Remove the last guide point
Clear	Reset the entire guide path
Build Road/River	Convert the guide path into a FlowyThing spline
Warn radius	Flags guide bends below the minimum safe radius

Note: Rivers usually need manual Y and rotation cleanup after Fit Terrain. Roads are a stronger match to terrain and need less manual work.

11. Mod Panel & Layers

The Mod panel is your main file management hub for mod projects.

Open Mod Folder	Load an existing mod. Must contain Definition.json.
Open Base Game	Load a standalone graph-data.json or game-graph.json directly.
New Mod	Creates a full mod folder structure. Prompts for mod ID and name.
Save All	Write all dirty layer files to disk. Same as Ctrl+S.
Layer visibility	Click the coloured dot to show/hide that layer on the map.
Active layer	Click the layer row to make it the write target for new edits.
● indicator	Layer has unsaved changes.
Ctrl+R	Reload all layers from disk. Yellow = unsaved changes will be lost.
Escape	Close the panel — mod stays loaded.

12. Progression Editor (Prog button)

The Progression editor manages the unlockable content chain in your mod — which track groups and map features become available as the player advances.

Left column	Sections in topological order (prerequisites first)	Section fields	ID, display name, prerequisites, purchase cost, feature to enable
Right column	Map features — what each section unlocks	Feature fields	ID, display name, areas unlocked, track group IDs
Click row	Select section or feature — details appear at the bottom	Prereqs	Comma-separated list of section IDs that must be purchased first
+ Section	Add a new purchasable section (dialog prompts for ID and name)	Track groups	GroupID values assigned to segments that this feature unlocks
+ Feature	Add a new map feature	Mousewheel	Scroll both the section and feature lists
Del	Remove the selected section or feature	Order	Sections are shown in dependency order automatically
Save	Write changes back to progressions.json		

Note: Track groups are linked to segments via the GroupID field in the segment properties panel. When a feature is unlocked in-game, all segments with matching GroupIDs become active.

13. Areas & Towns (Areas button)

The Areas panel edits the town and industry data that appears on the in-game map and controls where the AI routes deliveries.

Left column	Areas (towns) sorted by display order, with layer colour dot	Edit Area	Edit the full area JSON object in a text dialog
Middle column	Industries within the selected town	Edit Industry	Edit the full industry JSON in a text dialog
Right column	Components within the selected industry	Edit Comp	Edit the full component JSON in a text dialog
Click row	Select and edit — details appear in the field strip below	Save	Write all dirty town files back to disk
Go to Area	Pan the map to this area centre and close the panel	Del Area	Mark area for deletion (not saved until you click Save)
+ Area	Create a new area at the current map centre	Del Industry	Remove the selected industry from the area
+ Industry	Add an industry to the selected area	Del Comp	Remove the selected component from the industry
+ Component	Add a component to the selected industry	Mousewheel	Scroll any of the three columns

14. Scenery Placement (Scenery button)

Place static scenery objects — buildings, structures, props — by model ID anywhere on the terrain.

Model ID field	Type a Unity model path identifier, or click a quick-pick button	Object list	All placed scenery in the loaded layers with position details
Quick-picks	Common model IDs shown as one-click buttons below the field	Select row	Click a row to select that object
RotY nudge	±90 / ±45 buttons to rotate the object before placing	Go To	Pan map to the selected object
Place on Map	Enter placement mode — click map to drop at terrain height	Del Object	Remove the selected object from its layer
Escape	Exit placement mode without placing	Undo	Ctrl+Z undoes scenery placement
Y position	Auto-sampled from terrain at the click point	Writes to	Active mod layer, then bridge hot-reload

15. Mandela / Prefab Placement (Mandela button)

Mandelas are mod-placed instances of base-game prefabs — complex game objects like structures, vehicles, and special props that need to be referenced by their internal Unity prefab path rather than a simple model ID.

15.1 Setting Up the Prefab List

The editor reads available prefab paths from `dumped-mandelas.txt` — a catalog file from the Railroader game installation. The editor searches for it automatically in the default Steam location.

Browse ▼ button	Opens a searchable dropdown of all prefabs in the catalog. Type to filter. Press Enter or click to select.
Base Pick	Same as Browse — opens the full searchable list.
Reload Base	Reload the prefab catalog from disk.
Catalog status	Shows the source filename and prefab count, or an error if not found.
Current selection	The currently loaded prefab path is shown below the catalog status.

15.2 Placing a Prefab

1. Select prefab	Use Browse or Base Pick to choose a prefab path.
2. Set target path	Type a unique destination key in the Target field. Example: <code>MyMod/Structures/Depot_01</code> . Must be unique per mod.
3. Set Enabled	Cycle between Default / Force On / Disabled for this instance.
4. Place on Map	Enters placement mode. Click anywhere to drop at terrain height.
Escape	Exit placement mode without placing.

15.3 Managing Placed Mandelas

Placed list	All mandela instances in the loaded layers	Rotation X/Y/Z	Edit instance rotation in degrees directly in the draft panel
Click row	Select an instance — details load into the draft panel	Scale X/Y/Z	Edit instance scale directly
Load Sel	Pull the selected instance's values into the draft fields	Enabled mode	Default = game decides; Force On = always active; Disabled = always hidden
Save Sel	Write the draft field values back to the selected instance	Writes to	Owning mod JSON layer plus bridge hot-reload
Duplicate	Copy the selected instance with a new target path	Undo	Ctrl+Z works on mandela placement
Go To	Pan map to the selected instance's position		
Delete	Remove the selected instance from its layer		

16. Track Spans, Group Move & Calculators

16.1 Track Spans (Spans button)

Track spans define occupation detection sections — contiguous stretches of track bounded by specific segments at each end.

Span list	All spans across all loaded layers with their source layer colour
Click span	Select and edit upper segment, lower segment, distance, and end type
+ New Span	Create a new span entry
Del	Remove the selected span
Upper / Lower	The bounding segment IDs for this span. Click to type segment IDs.
Distance	Length of the span in metres

16.2 Group Move (Group button)

Select and move multiple nodes simultaneously.

Ctrl+drag map	Rubber-band box-select all nodes inside the rectangle
Shift+rubber-band	Add to the existing group selection
dX / dY / dZ	Translate the whole selection by this offset in metres
dRot (°)	Rotate the selection around its centroid by this many degrees
Apply	Commit the translation/rotation
Clear	Deselect all nodes in the group

Tip: Group move is ideal for repositioning an entire yard or station that was placed at the wrong coordinates.

16.3 Calculators (Calc button)

Four geometry calculators to help plan track layout without placing anything.

Crossover	Enter track separation and crossing angle. Outputs crossing length, leg length per side, and node count. Use this to space nodes correctly before hand-placing a diamond crossing.
Curved Turnout	Enter main-line radius, gauge, diverge angle, and desired leg length. Outputs diverge radius, arc length, lateral offset, and recommended minimum leg length. Use before configuring the Turnout tool.
Grade / Slope	Enter horizontal run and vertical rise. Outputs grade %, slope ratio, angle in degrees, and 3D track length. Also rates the grade as OK / Steep / Very Steep.
Measure	Click two nodes on the map to measure direct X/Z distance, 3D distance, delta Y, and average grade between them.

17. TrackBridge — Live Connection to the Game

TrackBridge is an optional Railroader mod (installed separately in-game) that creates a WebSocket connection between the editor and the running game. When connected, every track edit you make is reloaded in the game within ~500ms.

17.1 Connecting

●LIVE button	Green when connected, grey when not. The game must be running with the TrackBridge mod installed.
B key	Pick the Railroader game folder if the bridge cannot auto-detect it.
Auto-detect	The editor searches standard Steam paths for Railroader automatically at launch.
on_connect	Status bar shows 'Bridge: Railroader connected ●' when handshake succeeds.
on_disconnect	Status bar shows 'Bridge: Railroader disconnected' if the connection drops.

17.2 What the Bridge Does

Hot-reload	After every committed edit, the bridge calls <code>reload_tracks()</code> on the game. Track changes appear in-game without restarting.
Live positions	The map shows a live orange dot for the locomotive and teal dots for railcars.
Graph sync	The bridge can export the current in-game graph back to the editor for reference.

Note: Hot-reload only works for track graph changes. Terrain changes and progression changes require a full game restart to take effect in-game.

18. OSM Overlay (O key or OSM button)

The OSM (OpenStreetMap) overlay draws street map tiles over the terrain. Use it as a reference layer when placing roads, rivers, and track to match real geography.

O key	Toggle the OSM overlay on/off
OSM button	Same as O key, also opens opacity/zoom controls
Opacity	Blend strength of the map overlay (0 = invisible, 1 = full)
Zoom level	OSM tile zoom. Higher = more detail. Safe range is z10–z19.
Adjust zoom	Use the OSM panel controls to nudge the zoom up or down
Tile cache	Tiles are downloaded on first view and cached locally for reuse
Coordinate fit	The overlay is automatically georeferenced to match the tile coordinates

Tip: OSM zoom 15–17 is usually the best balance of detail and performance for route planning. Zoom 19 shows individual buildings and is useful for matching station locations.

19. Tile Generator (Generate button)

The tile generator downloads real-world elevation data from Mapbox and generates editor-compatible terrain tiles. You need a free Mapbox account and public API token to use this feature.

Token	Your Mapbox public access token. Get one free at mapbox.com/account . Click Paste to pull directly from the clipboard.
Output folder	Where generated .data tile files will be saved.
NLCD	Use NLCD land-cover data for vegetation (USA coverage only). Provides realistic biome distribution based on satellite data.
NLCD blur	Blur radius for smoothing biome boundaries.
Veg override	Force all generated tiles to use a single vegetation preset (0–7). Useful for non-USA areas where NLCD is unavailable.
Workers	Number of parallel download threads. Default 4. Higher = faster but more bandwidth.
Grid view	Click cells to queue them for generation. Drag to box-select multiple cells. RMB to dequeue.
Run	Start downloading and processing all queued tiles.
Progress	Each cell shows status: starting / downloading / processing / done / failed.
Presets	Save the full configuration (token, output, settings) as a named preset for quick switching between different projects.

Note: Generated tiles use the same RGBA PNG/.data format as hand-painted tiles. They can be mixed freely with hand-edited content.

20. Complete Keyboard Reference

View & Navigation

H	Heightmap display mode	Scroll	Zoom in / out
V	Vegetation display mode	LMB drag	Pan the map
W	Water display mode	MMB drag	Pan (middle mouse)
S	Toggle hillshade shading	RMB drag	Pan (right mouse)
D	Toggle diff overlay	F	Fit tiles in view
F	Fit all tiles in view	? / F1	Open in-app help
R	Redraw / refresh view	Esc / Q	Quit / cancel / close
T	Toggle track overlay	Ctrl+Z	Undo (terrain or mod)
N	Toggle node dots	Ctrl+S	Save all
O	Toggle OSM overlay	Ctrl+C	Copy selection
I	Toggle tile info tooltip	Ctrl+V	Paste selection
L	Load track graph JSON	Ctrl+X	Cut / export PNG
Ctrl+R	Reload from disk	← → arrows	Navigate help pages
Ctrl+-	Shrink UI scale	Tab	Also navigate help pages
Ctrl+=	Grow UI scale		
Ctrl+0	Reset UI scale to 100%		

Terrain Edit Mode (press E first)

E	Toggle edit mode	M	Toggle selection mode
B	Cycle brush type	0 - 7	Set veg preset (veg mode)
[]	Brush smaller / larger	,	Set height floor clamp
- =	Brush weaker / stronger	.	Set height ceiling clamp
MMB	Eyedropper — sample value	Ctrl+,	Clear floor clamp
LMB	Paint / raise	Ctrl+.	Clear ceiling clamp
RMB	Erase / lower	Ctrl+C	Copy selection
Ctrl+Z	Undo last stroke	Ctrl+V	Paste selection
Ctrl+S	Save all tiles	Ctrl+X	Cut / export
Ctrl+X	Export heightmap PNG	Delete	Clear selection

Track Editing

Ctrl+click map	Create node at cursor
Click node	Select node
Click segment	Select segment
Click again	Enter drag mode
Ctrl+click node	Connect to node
Drag → node	Snap connect
Drag → segment	Insert into segment
Shift+drag → seg	Insert + add turnout leg
Delete	Delete selected node/seg
Escape	Cancel drag / connect

Ctrl+Z	Undo mod edit (up to 50)
Ctrl+S	Save mod project
Ctrl+drag	Group rubber-band select
Shift+drag	Add to group selection
B	Pick Bridge game folder
Ctrl+R	Reload mod from disk
Ctrl+drag grid	Box-select generate cells
RMB grid cell	Dequeue generate cell
Mousewheel	Scroll list panels